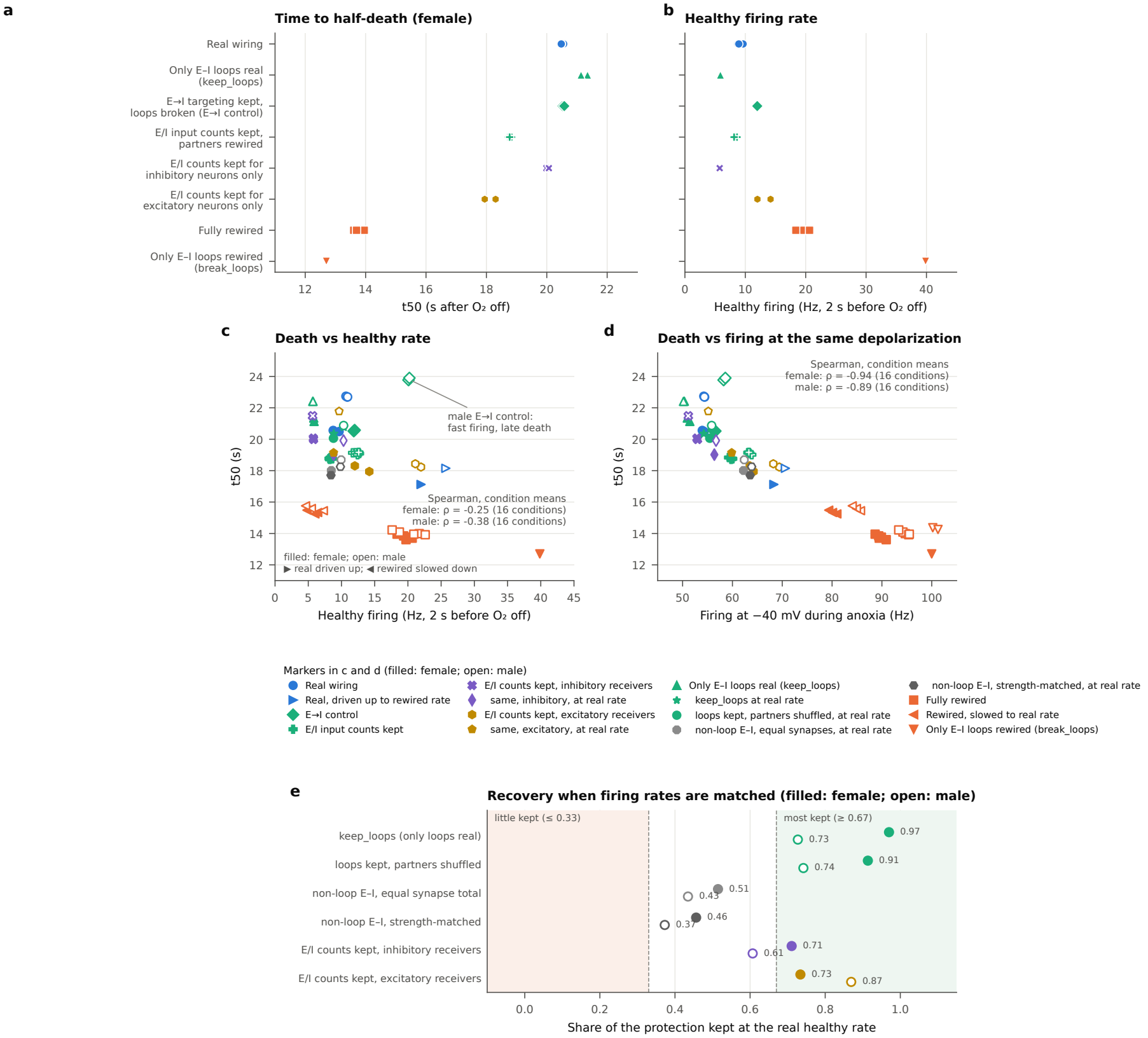
a
Time to half-death (female)
Real wiring
Only E–I loops real (keep_loops)
E→I targeting kept, loops broken (E→I control)
E/I input counts kept, partners rewired
E/I counts kept for inhibitory neurons only
E/I counts kept for excitatory neurons only
Fully rewired
Only E–I loops rewired (break_loops)
12 14 16 18 20 22
t50 (s after O₂ off)
b
Healthy firing rate
0 10 20 30 40
Healthy firing (Hz, 2 s before O₂ off)
c
Death vs healthy rate
t50 (s)
male E→I control: fast firing, late death
Spearman, condition means
female: ρ = -0.25 (16 conditions)
male: ρ = -0.38 (16 conditions)
filled: female; open: male
▶ real driven up; ◀ rewired slowed down
0 5 10 15 20 25 30 35 40 45
Healthy firing (Hz, 2 s before O₂ off)
d
Death vs firing at the same depolarization
Spearman, condition means
female: ρ = -0.94 (16 conditions)
male: ρ = -0.89 (16 conditions)
t50 (s)
50 60 70 80 90 100
Firing at −40 mV during anoxia (Hz)
Markers in c and d (filled: female; open: male)
Real wiring
Real, driven up to rewired rate
E→I control
E/I input counts kept
E/I counts kept, inhibitory receivers
same, inhibitory, at real rate
E/I counts kept, excitatory receivers
same, excitatory, at real rate
Only E–I loops real (keep_loops)
keep_loops at real rate
loops kept, partners shuffled, at real rate
non-loop E–I, equal synapses, at real rate
non-loop E–I, strength-matched, at real rate
Fully rewired
Rewired, slowed to real rate
Only E–I loops rewired (break_loops)
e
Recovery when firing rates are matched (filled: female; open: male)
little kept (≤ 0.33)
most kept (≥ 0.67)
keep_loops (only loops real)
loops kept, partners shuffled
non-loop E–I, equal synapse total
non-loop E–I, strength-matched
E/I counts kept, inhibitory receivers
E/I counts kept, excitatory receivers
0.73
0.97
0.74
0.91
0.43
0.51
0.37
0.46
0.61
0.71
0.73
0.87
0.0 0.2 0.4 0.6 0.8 1.0
Share of the protection kept at the real healthy rate